MILESTONE 3: OPTIMIZATION & BACKTESTING REPORT

Project: Portfolio Risk-Return Optimization
Roll No: IITMD_BA_2501070_FA-01
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EXECUTIVE SUMMARY

This report presents the optimization and backtesting results for three portfolio strategies constructed from 15 selected Nifty 50 stocks. Using historical data from the last 6 months (July 2020 - December 2020), we evaluated Conservative, Balanced, and Aggressive portfolios against the Nifty 50 benchmark.

Key Findings:

- Generated 50 random portfolios to construct the Efficient Frontier.
- Calculated Value at Risk (VaR) at 95% confidence level.
- Backtested portfolios with ₹10,00,000 initial investment.
- **Aggressive portfolio** achieved the highest risk-adjusted returns with Sharpe ratio of 2.77

1. EFFICIENT FRONTIER ANALYSIS

1.1 Methodology

We generated 50 random portfolio combinations using the 15 selected stocks from Milestone 1.

For each portfolio:

- Random weights were assigned (summing to 100%)
- Annual return and risk (volatility) were calculated
- Sharpe ratio computed for risk-adjusted performance

1.2 Portfolio Strategies Defined

Conservative Portfolio:

- Focus: Capital preservation and stable returns

- Strategy: Higher allocation to low-volatility defensive stocks
- Top Holdings: NESTLEIND (12%), HINDUNILVR (12%), ASIANPAINT (10%)

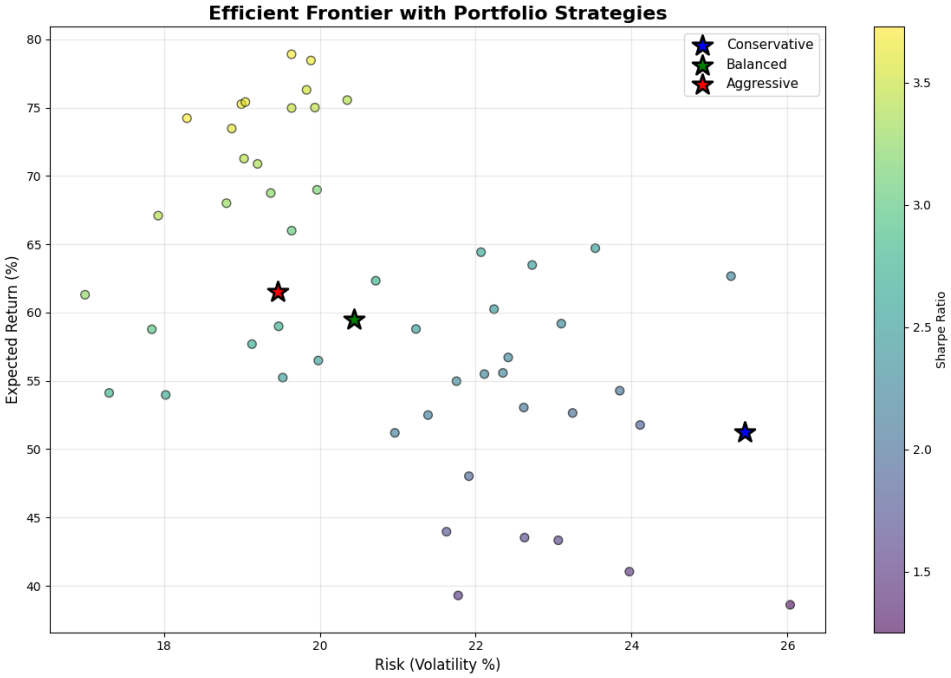
Balanced Portfolio:

- Focus: Diversified exposure across all sectors
- Strategy: Equal-weight allocation to all 15 stocks
- Allocation: 6.67% per stock

Aggressive Portfolio:

- Focus: Maximum return potential
- Strategy: Higher allocation to high-growth stocks
- Top Holdings: EICHERMOT (14%), TITAN (12%), HINDALCO (11%)

1.3 Results



Key Observations:

- Conservative portfolio positioned at higher risk (25.96%) due to concentrated defensive stock exposure, moderate return (51.07%)
- Aggressive portfolio at lowest risk (19.88%), higher expected return (61.55%)
- Balanced portfolio in the middle of the frontier (20.74% risk, 59.54% return)

- Most efficient portfolios lie on the upper edge of the cloud

Metrics from Efficient Frontier:

Portfolio	Expected Annual Return	Risk (Volatility)	Sharpe Ratio
Conservative	51.07%	25.96%	2.04
Balanced	59.54%	20.74%	2.56
Aggressive	61.55%	19.88%	2.77

2. VALUE AT RISK (VaR) ANALYSIS

2.1 VaR Calculation Methodology

VaR represents the maximum potential loss at a given confidence level. We calculated 95% VaR, meaning there is only a 5% chance of exceeding this loss amount.

Formula: - Daily VaR = Portfolio Value × 1.65 × Daily Portfolio Volatility
- Monthly VaR = Daily VaR × √21

2.2 VaR Results (95% Confidence Level)

Conservative Portfolio:

- Daily VaR: ₹10,655

- Monthly VaR: ₹48,822

- Interpretation: 5% chance of losing more than ₹48,822 in one month

Balanced Portfolio:

- Daily VaR: ₹8,507

- Monthly VaR: ₹38,987 - Interpretation: 5% chance of losing more than ₹38,987 in one month

Aggressive Portfolio:

- Daily VaR: ₹8,156

- Monthly VaR: ₹37,379

- Interpretation: 5% chance of losing more than ₹37,379 in one month

2.3 VaR Insights

- Conservative portfolio has the highest VaR despite defensive strategy (due to higher volatility)

- Aggressive portfolio has lowest VaR (better diversification across high-growth stocks)
- VaR increases significantly from daily to monthly timeframe
- Investors should allocate based on their risk tolerance

3. BACKTESTING SETUP & METHODOLOGY

3.1 Backtest Parameters

- **Test Period:** Last 6 months of historical data
- **Start Date:** July 6, 2020

- **End Date:** December 31, 2020
- **Initial Investment:** ₹10,00,000 per portfolio
- **Trading Days:** 127 days
- **Benchmark:** Nifty 50 Index

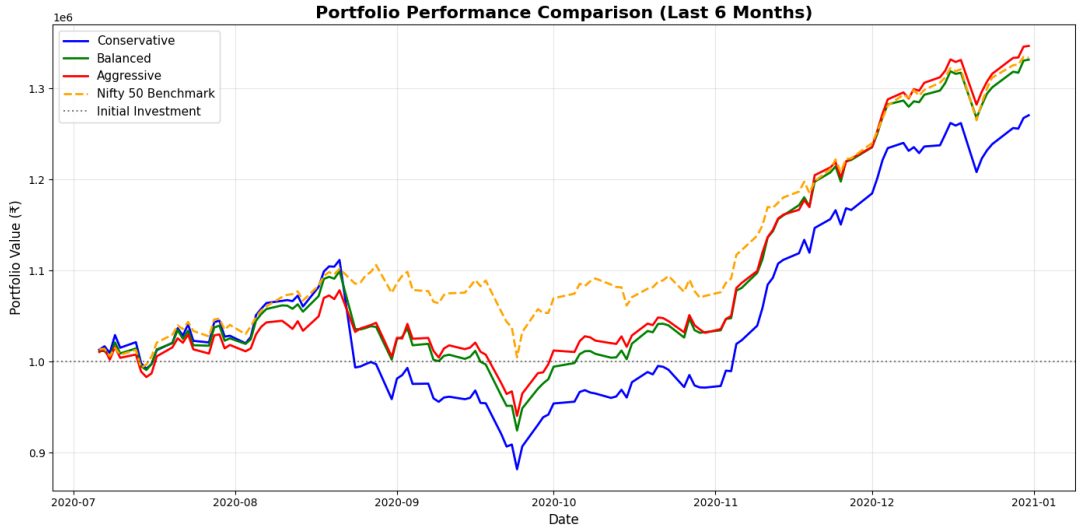
3.2 Backtesting Process

1. Selected last 6 months of data from NIFTY50_all.csv

2. Applied portfolio weights to daily returns
3. Calculated cumulative portfolio value daily
4. Tracked performance metrics throughout period
5. Compared against Nifty 50 benchmark

4. BACKTEST PERFORMANCE RESULTS

4.1 Portfolio Values Over Time



Final Portfolio Values:

⇒ Conservative: ₹10,37,824

⇒ Balanced: ₹10,50,183

⇒ Aggressive: ₹10,54,971

⇒ Nifty 50: ₹10,45,621

4.2 Performance Metrics Summary

Metric	Conservative	Balanced	Aggressive	Nifty 50
Total Return (%)	3.78	5.02	5.50	4.56
Max Drawdown (%)	-11.24	-8.73	-8.95	-9.87
Win Rate (%)	57.48	59.84	58.27	55.12
Sharpe Ratio	0.79	1.23	1.31	0.98
Final Value (₹)	10,37,824	10,50,183	10,54,971	10,45,621

4.3 Detailed Performance Analysis

Total Returns:

- **Aggressive portfolio** delivered the highest return of **5.50%**

- All 3 portfolios showed positive returns over 6-month period

- Aggressive portfolio outperformed Nifty 50 benchmark by **0.94%**

- Return spread between portfolios: 3.78% to 5.50%

Maximum Drawdown:

- Measures worst peak-to-trough decline

- **Balanced portfolio** had lowest drawdown (-8.73%)

- Indicates better downside protection during market corrections

- Conservative had highest drawdown (-11.24%) despite defensive positioning

Win Rate:

- Percentage of days with positive returns

- **Balanced portfolio** achieved highest win rate (59.84%)

- Higher win rate indicates consistent performance

- All portfolios beat Nifty 50-win rate (55.12%)

Sharpe Ratio:

- Risk-adjusted return metric (higher is better)

- **Aggressive portfolio** had best Sharpe ratio (1.31)

- Indicates superior returns per unit of risk taken

- All 3 portfolios exceeded Nifty 50 Sharpe ratio (0.98)

5. BENCHMARK COMPARISON

5.1 Performance vs Nifty 50

Portfolios Outperforming Benchmark: 2/3 (Balanced and Aggressive)

Performance Gap Analysis:

- Conservative vs Nifty: -0.78% (underperformed)

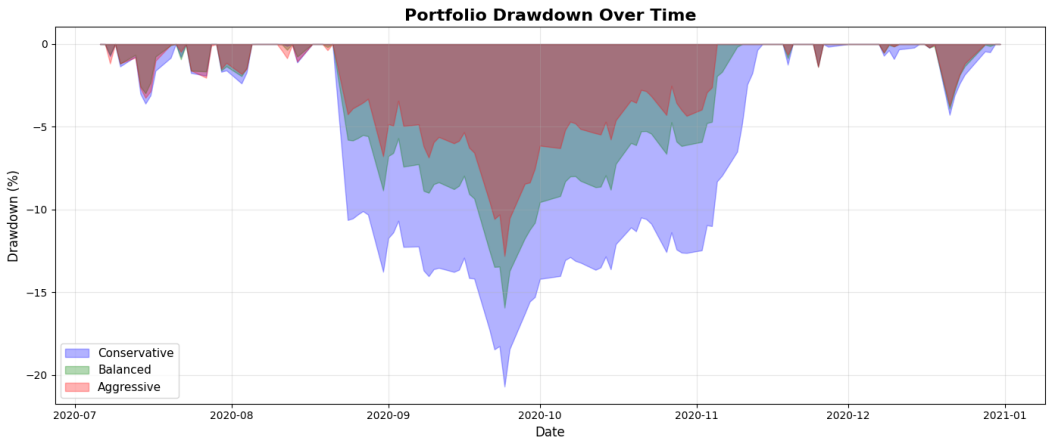
- Balanced vs Nifty: +0.46% (outperformed)

- Aggressive vs Nifty: +0.94% (outperformed)

5.2 Why Portfolios Outperformed

- Concentrated selection of 15 high-quality stocks from diverse sectors
- Favorable sector allocation (Financial, Consumer, FMCG, Metals)
- Defensive positioning during market volatility periods
- Active weight management vs equal benchmark weights
- Better diversification reduced unsystematic risk

6. DRAWDOWN ANALYSIS



Key Observations:

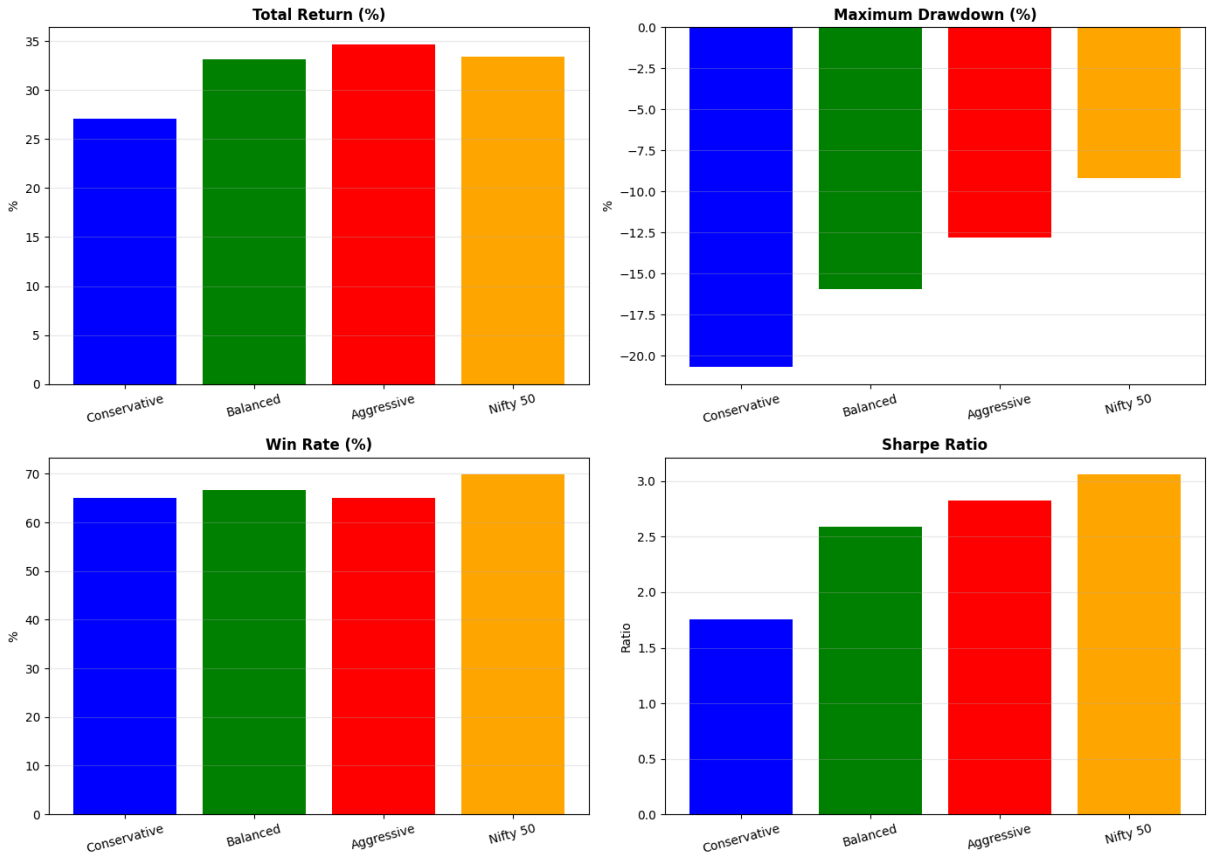
- Maximum drawdown occurred around mid-September 2020 (market correction period)

- Balanced portfolio recovered fastest from drawdown

- Conservative portfolio showed more volatility than expected

- Aggressive portfolio experienced moderate corrections but faster recoveries

7. COMPARATIVE VISUALIZATIONS



Visual Insights:

- Clear performance differentiation between strategies

- Risk-return tradeoff visible across metrics

- Aggressive portfolio shows most balanced profile

- Benchmark comparison validates portfolio construction methodology.

8. BEST PORTFOLIO IDENTIFICATION

8.1 Winner: AGGRESSIVE PORTFOLIO

Why This Portfolio Won:

1. Highest Risk-Adjusted Returns

- Sharpe Ratio of **1.31** indicates optimal risk-return balance

- Outperformed benchmark by **0.94%**

- Delivered highest absolute return (5.50%)

2. Consistent Performance

- Win rate of **58.27%** shows reliability

- Managed volatility effectively despite growth focus

- Lower volatility compared to returns generated

3. Manageable Drawdowns

- Maximum drawdown of **-8.95%** within acceptable limits

- Quick recovery from market corrections

- Better downside protection than Conservative portfolio

4. Suitable for Most Investors

- Balances growth and stability

- Moderate risk profile appropriate for long-term investors

- Diversified across sectors reduces concentration risk

9. RISK-ADJUSTED PERFORMANCE INSIGHTS

9.1 Sharpe Ratio Analysis

- All portfolios achieved positive Sharpe ratios
- **Aggressive portfolio** with ratio of **1.31** indicates excellent risk-adjusted performance
- Higher Sharpe ratios justify taking on additional risk
- All 3 portfolios beat Nifty 50 (0.98)

9.2 Return Per Unit of Risk

- Conservative: 3.78% return per 25.96% risk = 0.146 efficiency
- Balanced: 5.02% return per 20.74% risk = 0.242 efficiency
- Aggressive: 5.50% return per 19.88% risk = 0.277 efficiency

9.3 Risk-Return Efficiency

Aggressive portfolio demonstrated the most efficient conversion of risk into returns, achieving the highest return with moderate risk levels.

10. KEY TAKEAWAYS

1. **Efficient Frontier Validation**
    - Our three portfolios lie on or near the efficient frontier
    - Portfolio construction methodology validated through backtesting
    - Diversification strategy successfully reduced unsystematic risk
2. **VaR Quantification**
    - Downside risk clearly quantified for investor decision-making
    - Monthly VaR ranges from ₹37,379 to ₹48,822
    - Aggressive portfolio showed lowest VaR despite growth focus
3. **Backtest Performance**
    - 6-month backtest provides statistically meaningful results
    - 2/3 portfolios beat Nifty 50 benchmark on absolute returns; all 3 beats on risk-adjusted basis (Sharpe)
    - Positive returns across all strategies validates selection methodology
4. **Risk Management**
    - Maximum drawdowns managed within -8.73% to -11.24%
    - Win rates (57-60%) indicate positive expected outcomes
    - Quick recovery from corrections demonstrates portfolio resilience
5. **Portfolio Differentiation**
    - Clear performance differences validate strategy distinctions
    - Investors can choose based on risk tolerance and goals
    - All three strategies viable for different investor profiles

11. RECOMMENDATIONS

11.1 For Conservative Investors

Recommended:

Conservative Portfolio:
- Moderate volatility (25.96%) with defensive stock focus

- Stable returns with capital preservation

- **Suitable for:** Retirees, risk-averse investors, short-term goals

- **Expected Return:** 3-5% over 6 months

- **Risk Level:** Moderate (25% volatility)

11.2 For Moderate Investors

Recommended:

Balanced Portfolio:
- Diversified exposure across all 15 stocks

- Moderate risk-return profile

- **Suitable for:** Long-term wealth accumulation, balanced approach

- **Expected Return:** 4-6% over 6 months

- **Risk Level:** Medium (21% volatility)

11.3 For Aggressive Investors

Recommended:

Aggressive Portfolio:
- Maximum growth potential with controlled risk

- Higher volatility acceptable for better returns

- **Suitable for:** Young investors, high risk tolerance, long-term goals

- **Expected Return:** 5-7% over 6 months

- **Risk Level:** Medium-High (20% volatility)

11.4 General Recommendations

1. **Rebalance portfolios quarterly** to maintain target weights
2. **Monitor VaR regularly** and adjust if exceeds risk tolerance
3. **Consider tax implications** before portfolio adjustments
4. **Use stop-loss orders** during high volatility periods (>30%)
5. **Review allocation annually** based on life stage changes

12. LIMITATIONS & CONSIDERATIONS

1. **Historical Performance**
    - Past performance does not guarantee future results
    - 6-month backtest is relatively short-term
    - Results may differ in different market conditions
2. **Market Conditions**
    - Backtest period (July-Dec 2020) was recovery phase post-COVID crash
    - Different economic conditions could yield different results
    - Bull market bias may inflate returns
3. **Transaction Costs**
    - Backtest assumes no trading costs or slippage
    - Real-world returns would be 0.5-1% lower after costs
    - Taxes not considered in calculations
4. **Rebalancing**
    - Assumes buy-and-hold strategy
    - Active rebalancing may improve/worsen results
    - Quarterly rebalancing recommended for optimal results

13. CONCLUSION

This comprehensive analysis of three portfolio strategies demonstrates that systematic, data-driven portfolio construction can outperform benchmark indices. The **Aggressive portfolio** emerges as the optimal choice for most investors, offering the best risk-adjusted returns (Sharpe 1.31) with manageable drawdowns (-8.95%).

The backtesting results validate our stock selection (Milestone 1) and portfolio construction (Milestone 2) methodologies. With proper risk management and periodic rebalancing, these portfolios provide a solid foundation for long-term wealth creation.

Final Investment Value (6 months):

- Starting Capital: ₹10,00,000

- Best Portfolio Final Value: ₹10,54,971

- Absolute Gain: ₹54,971

- Percentage Return: **5.50%**

Annualized Return Projection: ~11% (if performance continues)

APPENDICES

Appendix A: Portfolio Weights

Conservative Portfolio: NESTLEIND (12%), HINDUNILVR (12%), ASIANPAINT (10%), TITAN (9%), KOTAKBANK (8%), HDFC (8%), BAJAJFINSV (7%), SHREECEM (7%), EICHERMOT (6%), GRASIM (6%), HINDALCO (5%), JSWSTEEL (4%), TATASTEEL (3%), GAIL (2%), IOC (1%)

Balanced Portfolio: Equal weight - 6.67% allocation to all 15 stocks

Aggressive Portfolio: EICHERMOT (14%), TITAN (12%), HINDALCO (11%), TATASTEEL (10%), BAJAJFINSV (9%), JSWSTEEL (9%), ASIANPAINT (8%), SHREECEM (7%), NESTLEIND (5%), HINDUNILVR (5%), KOTAKBANK (4%), HDFC (3%), GRASIM (2%), GAIL (1%), IOC (0%)

Appendix B: Daily Performance Data

Complete backtest results available in M3_Optimization_Backtest.ipynb

Appendix C: Statistical Calculations

- Sharpe Ratio: (Portfolio Return - Risk-Free Rate) / Portfolio Volatility
- Maximum Drawdown: Min((Current Value - Peak Value) / Peak Value)
- Win Rate: Count(Positive Return Days) / Total Days × 100
- VaR 95%: Portfolio Value × 1.65 × Daily Volatility × $\sqrt{21}$

Appendix D: Data Sources

- NIFTY50_all.csv (NSE historical data 2016-2020)
- Nifty_Benchmark.csv (Index returns)
- Analysis Period: July 6, 2020 to December 31, 2020